

Bootstrapping:

- 1) Randomly sample subset of observed data.
- 2) Calc. mean of random sample
- 3) Repeat 1+2 until you have many means

95% confidence interval - Interval that spans 95% of bootstrapped means.

□ p-value of anything outside of confidence interval is $< 0.05 \Rightarrow$ statistical significance.

t-test - Test of significance of difference b/t two independent samples from the same population.

Confidence interval for normally-distr. data:

$$\underline{CI = \bar{x} \pm z \cdot \frac{s}{\sqrt{n}}}$$

Confidence interval for binomial data - For:

$i \in \{1, \dots, n\}$, $n = 450$, and assuming $C_i \stackrel{iid}{\sim} \text{Bern}(p)$

$$\underline{X = \sum_i c_i \sim \text{Bin}(n, p), \hat{p} = \frac{X}{n}, \text{Var}(\hat{p}) = \frac{p(1-p)}{n}}$$

in this case, p is fixed, $\hat{p} = \frac{X}{n}$ is an estimator, i.e. RV.

$$\underline{\text{Wald Statistic} - Z = \frac{\hat{\beta} - \beta_0}{\text{se}}}, \text{ for}$$

$$\text{se} = \sqrt{p(1-p)/n}$$

Wald Confidence Interval - Set of β_0 values for which:

$$|Z| < 1.96 \text{ (95 \% confidence)}$$

Ex. of test using Wald for Binomial Param:

$$H_0: \pi = 0.50, H_a: \pi \neq 0.50, \hat{p} = 0.90 \text{ for } n=10;$$

Wald test:

$$SE = \sqrt{0.9(0.1)/10} = 0.095$$

$$\Rightarrow Z = (0.9 - 0.5)/0.095 = 4.22$$

$$\Rightarrow p < 0.001$$

Sample Size for Comparing Two Proportions

To determine sample size to test whether "success" probs of two groups π_1 and π_2 are identical, we must specify prob. β_0 of failing to detect difference between groups.

$$\square \beta_0 = P(\text{rejecting } H_0 \text{ at level } \alpha) \text{ for } p\text{-val} \leq \alpha.$$

$$\alpha = P(\text{Type I error}), \beta = P(\text{Type II error}).$$

$$\square \text{Power} = 1 - \beta$$

Models for Matched Pairs

Two-way contingency table — A summary of matched-pair observations displaying category counts for both classifications.

□ Ex — Opinions Relating to Environment

pay "yes" or no taxes	Cut Living Standards		Total
	yes	no	
yes	227	132	359
no	107	676	785
Total	334	810	1144

For $\pi_{ijj} = P(\text{outcome } i \text{ for } Q1, \text{ outcome } j \text{ for } Q2) :$

□ Marginal homogeneity — $\pi_{1+} - \pi_{+1} = \pi_{12} - \pi_{21}$

□ Because samples are from similar populations (i.e. opinions of same subjects on questions regarding environment), $n_{2+} \approx n_{+1}$, i.e. the # of people answering yes to either question is similar (359 vs. 334).

$\Rightarrow$ marginal populations are correlated.

□ Hence, best way to test difference in populations is to test existence of marginal homogeneity.

$H_0 : \pi_{1+} = \pi_{+1}$, or $H_0 : \pi_{12} = \pi_{21}$

Estimating Differences of Proportions

Let $p_{i,j} = n_{i,j}/n$,

$$\mathbb{E}[p_{1+} - p_{+2}] = \pi_{1+} - \pi_{+2} ;$$

$$\text{Var}(p_{1+} - p_{+2}) :$$

$$[p_{1+}(1-p_{1+}) + p_{+1}(1-p_{1+}) - 2(p_{11}p_{22} - p_{12}p_{21})]/n$$

with $SE = \sqrt{\text{Var}(p_{1+} - p_{+1})}$

$$= \sqrt{(n_{12} + n_{21}) - (n_{12} - n_{21})^2/n} / n$$

Review Problems

All problems from Agresti "An Intro to Categorical Data Analysis"

1.8) a) $n = 1170$, $\hat{p} = \frac{344}{1170} = 0.294$

b) $H_0: \pi = 0.5$, $H_A: \pi \neq 0.5$:

$$SE = \sqrt{\hat{p}(1-\hat{p})/n} = \sqrt{(0.294)(0.706)/1170}$$

$$= 0.0133$$

$$\Rightarrow z = \frac{(0.294) - (0.5)}{0.0133} = -15.49$$

$$\Rightarrow p < 0.001$$

A p -value of < 0.001 suggests that, with 95% confidence, we can say that the true proportion of people willing to accept cuts in their standard of living is not 0.5.

c) For 99% confidence, $z = 2.576$:

$$\Rightarrow \text{Half interval (HI)} = 2.576(0.0133)$$

$$= 0.0343$$

$$\Rightarrow p \in [0.466, 0.5343]$$

8.1)

Pay higher taxes	Cut Living Standards		Total
	yes	no	
yes	227	132	359
no	107	676	785
Total	334	810	1144

$$Z = \frac{n_{12} - n_{21}}{\sqrt{n_{12} + n_{21}}} \approx \frac{132 - 107}{\sqrt{132 + 107}}$$

$$= \frac{25}{\sqrt{239}} \approx 1.62$$

$$\Rightarrow p = 0.106 \neq 0.05$$

$\Rightarrow$ The difference between $\hat{\pi}_{12}$ and $\hat{\pi}_{21}$ is not significant, meaning their

is no evidence to suggest a difference in the population of those who believe in cutting living standards v.s. paying higher taxes to preserve the environment.

$$8.2) a) H_0: \pi_{1+} = \pi_{+1}, H_A: \pi_{1+} \neq \pi_{+1}$$

$$Z = \frac{n_{12} - n_{21}}{\sqrt{n_{12} + n_{21}}} = \frac{125 - 2}{\sqrt{125 + 2}} \approx 10.914$$

$\Rightarrow (p < 0.001 \Rightarrow)$ Reject null hypothesis

$$b) SE = \sqrt{(n_{12} + n_{21}) - (n_{12} - n_{21})^2 / n} / n$$

$$= \sqrt{(125 + 2) - (125 - 2)^2 / 1120} / 1120$$

$$= 0.00951, \quad p_{1+} - p_{+1} = \frac{958}{1120} - \frac{835}{1120}$$

$$= 0.110$$

$$\Rightarrow 0.110 \pm 1.645(0.00951) = [0.0944, 0.1256]$$

We can say with 90% confidence that the true difference in the proportion of those that believe in heaven v.s. those that believe in hell is within $[0.0944, 0.1256]$

8.4) Both tests are meant to examine whether a difference exists in populations that are represented using two dependent samples.